

CHE 313

1. Course number and name

CHE 313. Transport phenomena III

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Sangil Kim, Christos Takoudis, Gang Cheng

4. Textbook, title, author, publisher, year

Wankat, P. C., Separation Processes Engineering: Includes Mass Transfer Analysis, 4th edition, Prentice Hall, 2017. -OR-

Christie John Geankoplis, Allen Hersel, Daniel H. Lepek, Transport processes and Separation Process Principles, 5th edition, Prentice Hall, 2018

a. Other supplemental materials

Seader, J.D et al., Separation Processes Principles, 3rd edition, John Wiley & Sons, New Jersey, 2011.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Mass transfer and phase equilibria. Multistage separations; applications in distillation; extraction; absorption and drying.

b. Prerequisites or co-requisites

CHE 301

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Be familiar with separation principles, design, separation mechanisms, processes, and applications
- Understand the underlying thermodynamic and mass transfer principles
- Understand the operation of simple and multistage batch distillation systems, and McCabe-Thiele method
- Discuss the differences between batch and continuous operation
- Explore use of various types of separation techniques
- Explore use of alternative separation techniques to the existing ones

- a. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

Introduction to separation processes
Flash Distillation
Binary distillation
Multicomponent distillation
Batch distillation
Stage and packed column design
Absorption and stripping
Liquid-liquid extraction
Membrane separations